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A Overview of the Appendix

In the appendix, we provide additional details and supplementary information to further elaborate on sections mentioned above. In Section B, we analyze the statistical features of the dataset, meanwhile providing examples of questions ranging from different categories. Section C contains experiments details, including the evaluation of LLMs, the evaluation of hint prompts and RL experiments. Some examples of model outputs are also illustrated.

B Benchmark Analysis

B.1 Statistical analysis

As shown in Figure 10, the text length of questions in VisuLogic is mostly concentrated around 40 tokens (calculated by Llama-3.1's and InternVL2.5's tokenizer). We also analyze the distribution of image sizes, as shown in Figure 9. The image widths range from 200 to 700 pixels, with an average of 592.3 pixels, while the heights range from 90 to 825 pixels, with an average of 327.9 pixels.

Figure 9: **Image size distribution.** The size of images is limited to within the same order of magnitude.

B.2 More Examples of VisuLogic

To provide a thoroughly presentation of our benchmark, we include more examples of questions from different categories in the Figure 11 and Figure 12.

C Evaluation & Experiment

C.1 Evaluation of LLMs

Caption generation for LLMs Evaluation. In our experiment, we employ large language models (LLMs) for comparative analysis. Specifically, when setting up the LLM-based experiment, we initially utilize GPT-4o to generate captions for images with the following prompt: *Please describe the fine-grained content of the image or figure based on this question, including scenes, objects, relationships, and any text present. Please note that you do not need to answer this question directly, just describe the information of this picture*. Additional examples of generated image captions are presented in Figure 14 and Figure 15.

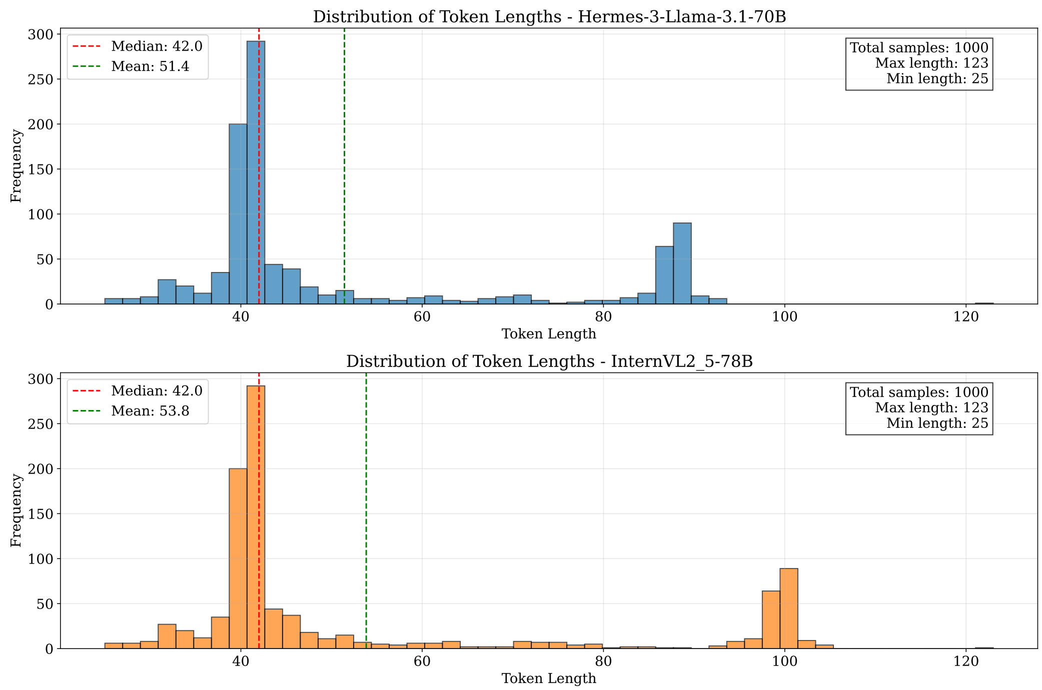

Figure 10: Distribution of text token length in VisuLogic.

More Examples of Captions. We provide additional image captions for six categories, as illustrated in Figures 14 and 15. Even SOTA MLLM (GPT-4o) encounters difficulties in accurately describing the details of images from VisuLogic.

C.2 More Solutions from Models

We provide more solutions generated from different LLMs/MLLMs on our benchmark, as shown in Figure 16, Figure 17 and Figure 18. For the majority of questions, almost all models fail to provide accurate solutions. Sometimes even when the final answer is correct, methodological wrong may persist.

C.3 Hint Prompts Evaluation Details

We first generate hint prompts with GPT-4o, combining reference solutions with question data as inputs. All outputs undergo manual validation to prevent solution leakage. More examples are shown in Figure 19. After that, we input the hint prompts along with the same CoT prompt in CoT experiments ("*Solve the complex visual logical reasoning problem through step-by-step reasoning. Think about the reasoning process first and answer the question following this format: Answer: \boxed{$LETTER}*.") to MLLMs.

C.4 RL Experiments

Comparative SFT Experiments. To verify the effectiveness of RL method, we arrange the comparative SFT experiments on the same dataset as RL experiments. The instruction consists of questions and Non-CoT prompts, and the responses are formatted direct answers.

RL Algorithm. We employ REINFORCE Leave-One-Out (RLOO) [3] in our reinforcement learning training phase. As a critic-model-free algorithm,rloo is at a low computational cost while maintaining more robustness to noise and KL constraints.